

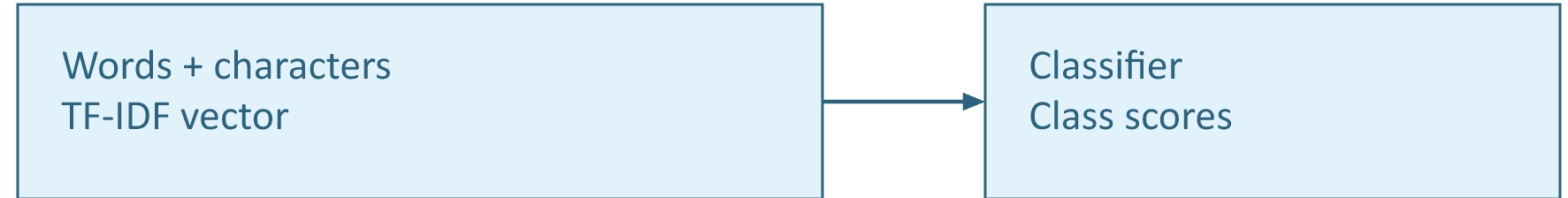
Violence report classification

Model selection, Severe augmentation
and final decisions around an anchor

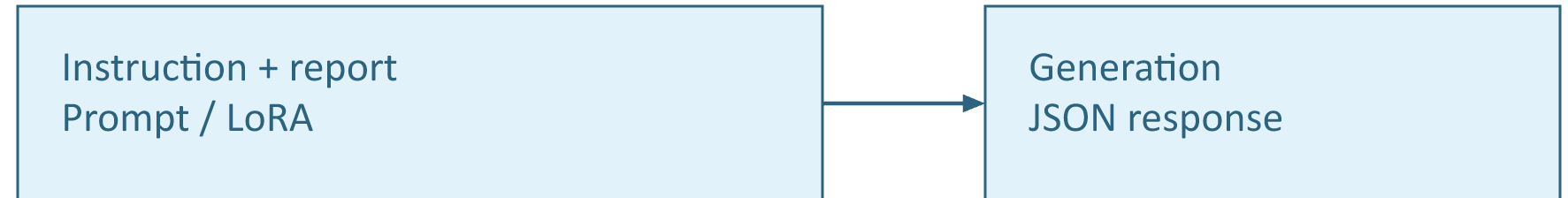
Marcel Herrera Rendón
Universidad de Sonora

Three families, three classification paths

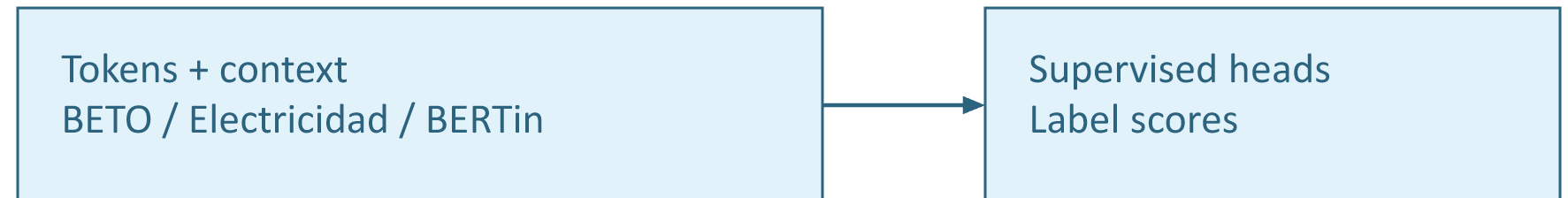
Classical



Generative LLMs



Transformer encoders

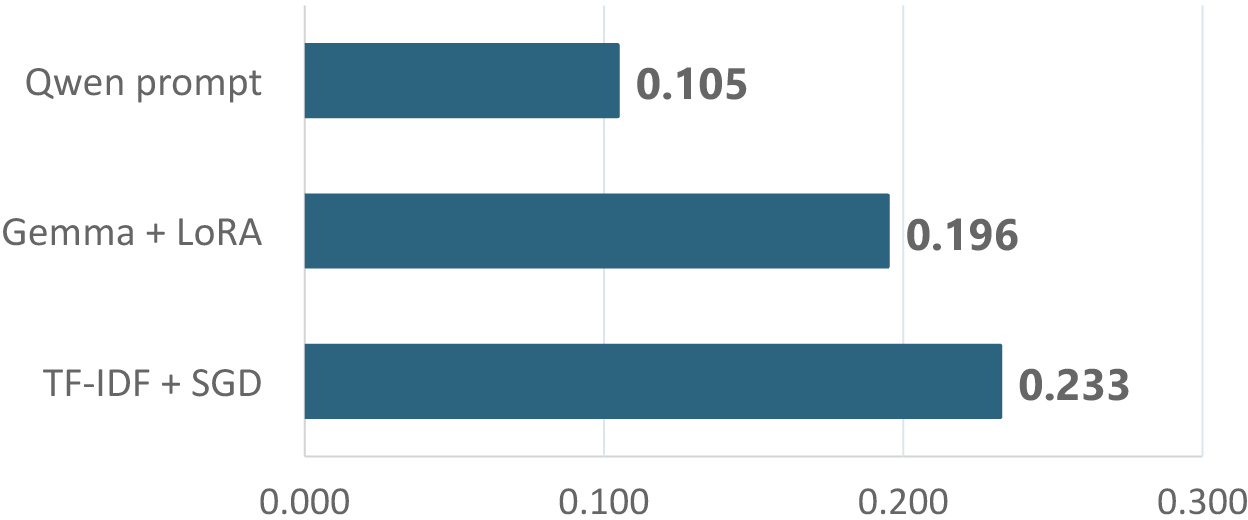


LLMs also use Transformers. Here we distinguish generative and discriminative use.

Evidence guided model selection

Pilots on the same 133 reports

Joint exact match S1 + S2



LoRA pilot: 625 training examples and 80 adaptation steps.

Classical models versus BETO

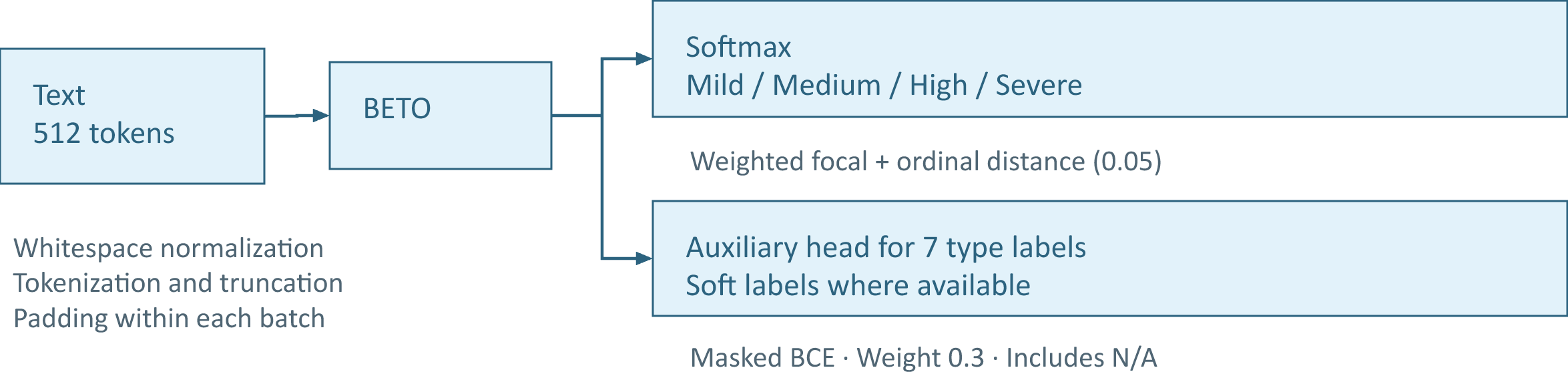
Full development

Metric	Classical	BETO
S1 macro-F1	0.4974*	0.5429
S2 macro-F1	0.7642	0.8198
S2 micro-F1	0.8449	0.8685

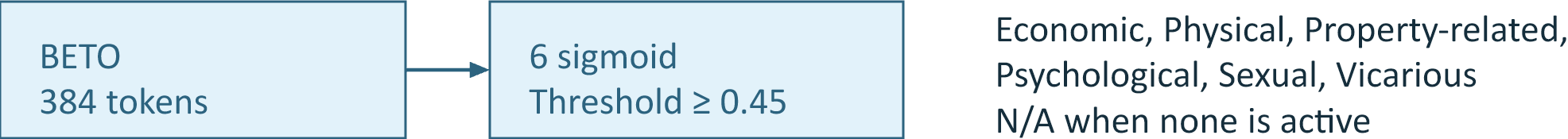
*S1: early classical reference.
S2: character SVM / BETO threshold 0.45.

The problem determines the outputs

S1 · Augmented BETO recipe (augrep08)

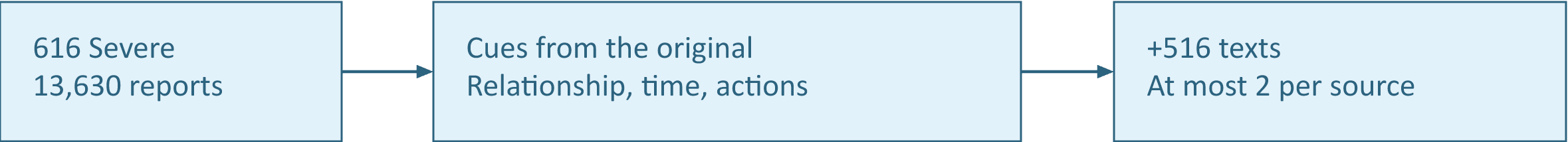


S2 · Separate final component



Targeted Severe augmentation

Training only · Originals retained · Labels inherited

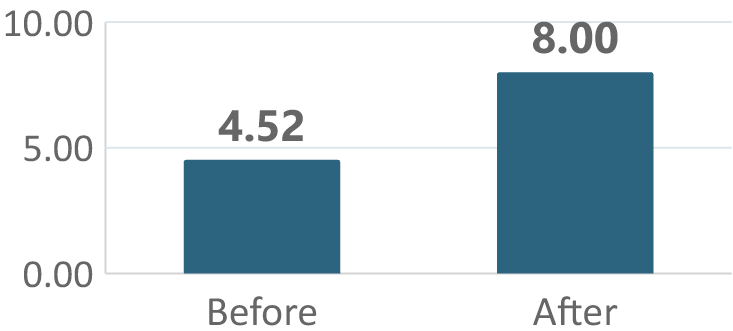


Composition using report templates



Stable seed per record
Cue-conditioned composition
Template-based generation

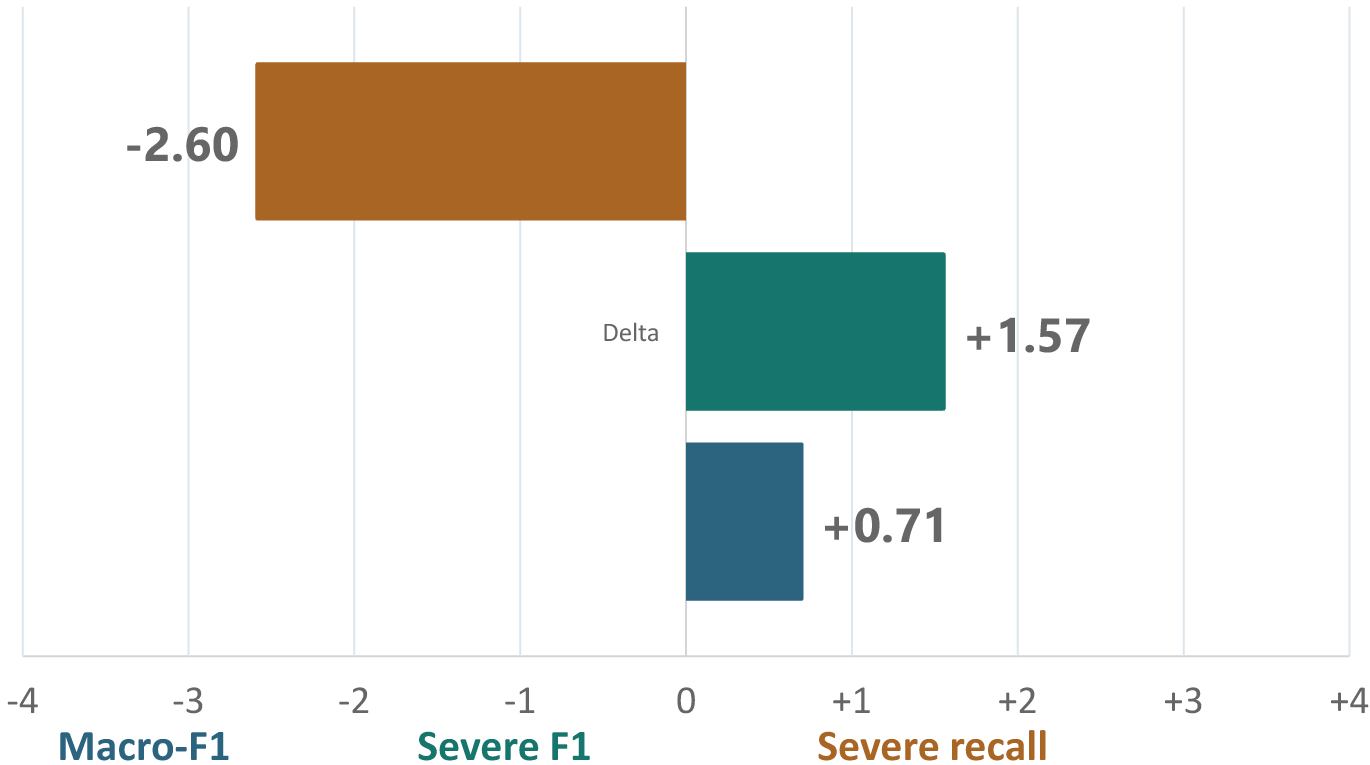
Severe share (%)



1,132 / 14,146 · Partial rebalancing

The improvement includes an error trade-off

Absolute change × 100 · Zero = no change



Same 3,405 development reports.
No augmentation at evaluation.

Metric	Base	Aug.
Macro-F1	0.5481	0.5552
Severe F1	0.4178	0.4335
Recall	0.3961	0.3701

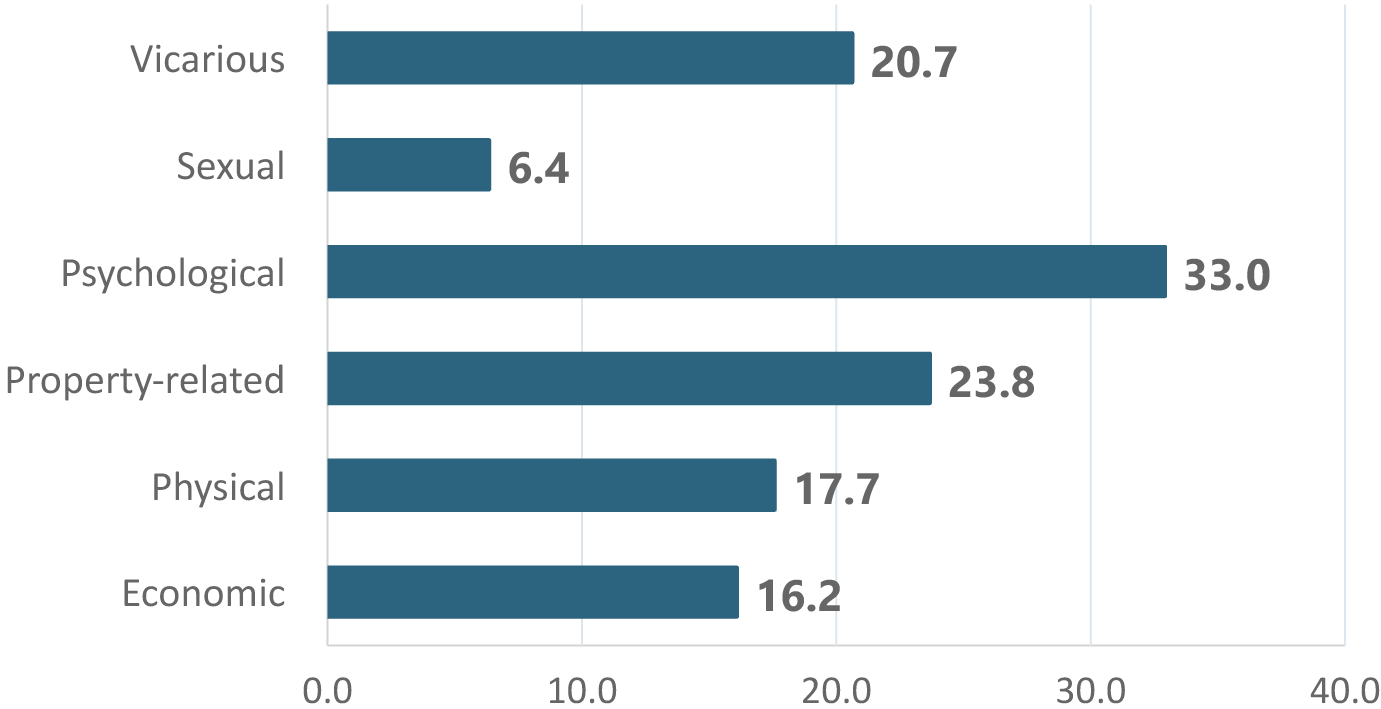
F1 rises; recall falls

Class weights are recomputed
on the augmented training set.

Disagreement remains in the labels

New analysis of S2 soft labels · Development: 1,212 reports

Split-vote decisions (%) · Denominator: all reports



Three votes per type decision; soft labels retain their proportion.

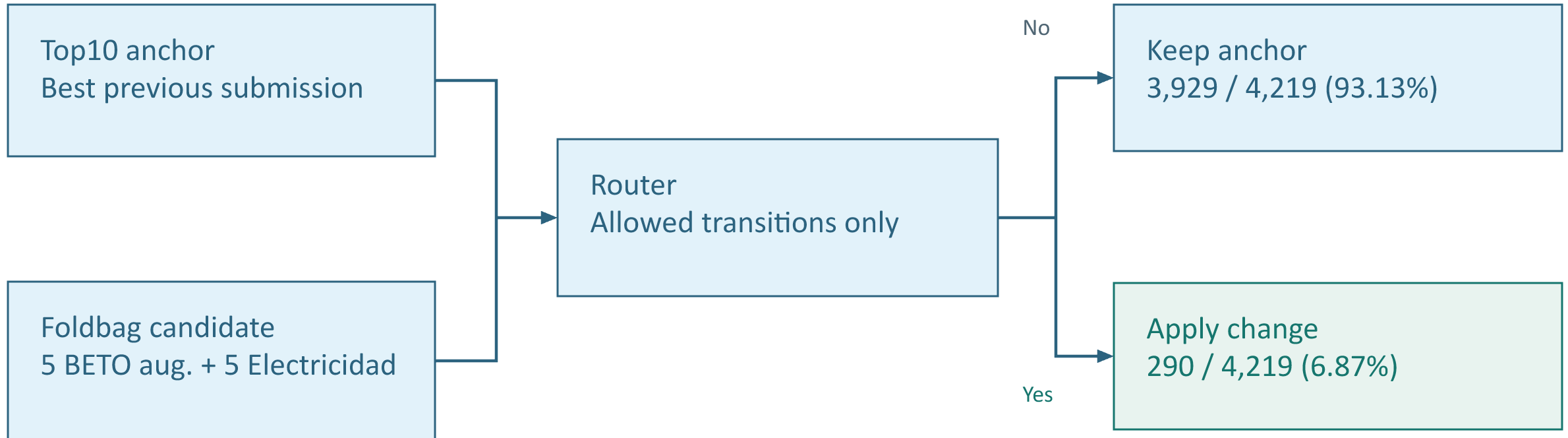
72.36%

with at least one split type
877 of 1,212 reports



Example of 3 votes.
Within-item variance: $p(1-p) = 2/9$.

The anchor limits final changes



Medium → Mild: 177 Mild → Medium: 101 High → Severe: 12

High → Severe requires severe lexical cues. Selected recipes + a final decision rule.

Complete system results

SEVERITY · MACRO-F1

0.60798

3rd / 16 teams

Anchor 0.60678 → Final 0.60798
Change: +0.00120

TYPES · MICRO-F1

0.87016

6th / 15 teams

S2 predictions
unchanged in final step

Official test scores informed submission selection.

A strategy to test in other settings

Compare families under the same protocol
Match outputs and losses to the labels
Evaluate augmentation by class and error type

Next tests

Variant audit · Controlled balance and weights
Repeated seeds · Validation by institution and consensus

Code and evidence: github.com/mariron42/clasificador_violencia

The augmented model in an ensemble

Intermediate ensemble, May 5, 2026

Original development: 3,405 reports, including 154 Severe

System	Macro-F1	Severe F1	Severe recall
Baseline BETO	0.5481	0.4178	39.61%
BETO + Severe augmentation	0.5552	0.4335	37.01%
Ensemble with augmented BETO	0.5805	0.4737	40.91%

Severe recall exceeds the baseline

10 equally weighted sources, including BETO with Severe augmentation.

Backup: complete ensemble result on development